

Electromagnetic Theory

Some Basic Formulae:

UNIT-I

(1)

- Most of the physical quantities existing in physics are either scalar or vector. A scalar quantity has magnitude only. It can be added, subtracted, multiplied or divided by the simple rules of arithmetic.
- A vector quantity has both magnitude and direction and can be combined according to the certain rules of addition.
- The product of two vectors may be scalar or vector. If the product of two vectors is a scalar quantity, then it is called scalar product ($\vec{A} \cdot \vec{B}$) [read $\vec{A}$ dot $\vec{B}$] and if the product is vector quantity, then it is called vector product ($\vec{A} \times \vec{B}$) [read $\vec{A}$ cross $\vec{B}$]

● Product of Two Vectors:

- (i) Dot OR Scalar Product: $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$

The scalar product obeys the commutative law, i.e. $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

- (ii) If vectors $\vec{A}$ and $\vec{B}$ are orthogonal (i.e. perpendicular to each other),

$$\theta = 90^\circ \text{ and } \cos 90^\circ = 0 \quad \therefore \vec{A} \cdot \vec{B} = AB \cos 90^\circ = 0$$

$$\text{Similarly } \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

- (iii) If vectors are parallel, $\theta = 0^\circ$ and $\cos 0^\circ = 1 \quad \therefore \vec{A} \cdot \vec{A} = AA = A^2$

$$\text{Similarly } \hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

- (iv) If $\vec{A} = \hat{i}A_x + \hat{j}A_y + \hat{k}A_z$ and $\vec{B} = \hat{i}B_x + \hat{j}B_y + \hat{k}B_z$, then

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

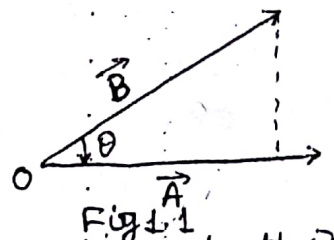


Fig. 1.1

● Cross OR Vector Product: The cross product

of two vectors $\vec{A}$ and $\vec{B}$ is expressed as

$$\vec{A} \times \vec{B} = \vec{V}, \text{ where } \vec{V} = |\vec{A}| |\vec{B}| \sin(\angle \vec{A}, \vec{B}) \hat{n}$$

$$\vec{V} = \vec{A} \times \vec{B} = |\vec{A} \times \vec{B}| \hat{n}$$

Here $\hat{n}$ is the unit vector normal to the plane containing $\vec{A}$ and $\vec{B}$ as shown in Fig. 1.2.

- If vectors $\vec{A}$ and $\vec{B}$ are perpendicular to each other, i.e. $\theta = 90^\circ$, $\sin 90^\circ = 1$

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \hat{n}$$

$$\text{Thus, } \hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j} \quad \text{and} \\ \hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = -\hat{i}, \hat{i} \times \hat{k} = -\hat{j}$$

If the vectors $\vec{A}$ and $\vec{B}$ are parallel, i.e. $\theta = 0^\circ$, $\sin 0^\circ = 0 \quad \therefore \vec{A} \times \vec{B} = 0$

$$\text{Thus } \vec{A} \times \vec{A} = \vec{B} \times \vec{B} = \vec{C} \times \vec{C} = 0, \text{ Similarly } \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

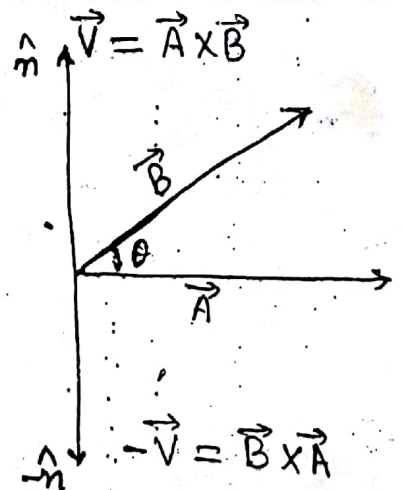


Fig. 1.2

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●● Gradient of a Scalar Function: The gradient of a continuously differentiable scalar function $\phi(x, y, z)$ is mathematically defined as

The gradient of a scalar field is a vector field.

$$\text{grad } \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \phi = \vec{\nabla} \phi$$

= a vector operator

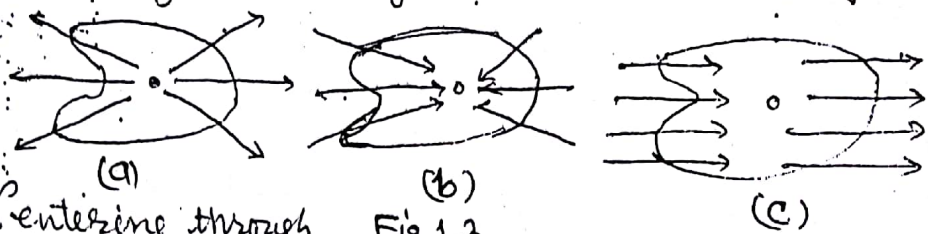
where; $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$. Here, $\vec{\nabla}$ is called the vector operator or del operator and $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along x, y, z axes respectively. In fact a derivative tells us how fast the function varies.

●● Divergence of a Vector Function $\vec{F}(x, y, z)$: The divergence of a vector field at a given point in a vector field is a scalar field and is defined as the amount of the flux diverging from unit volume element per second around the point. Mathematically, the divergence of a vector $\vec{A}$ is defined as the scalar product of the vector operator $\vec{\nabla}$ and the vector $\vec{A}$, i.e.,

$$\text{div } \vec{A} = \vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Physical Significance of Divergence: Divergence of a vector field at a point is a measure of flux of that vector field at that point. If the divergence of a vector field at a point is +ve, it means that the flux is coming out through that surface [Fig. 1.3(a)]

If the divergence of a vector field is -ve, it means



that the flux is entering through the surface [Fig. 1.3(b)]

If the divergence of a vector field is zero means that the flux entering through the surface is equal to the flux leaving through that surface as shown in Fig. 1.3(c).

●● Curl of a Vector Field: The curl of a vector field at any point is defined as vector quantity whose magnitude is equal to maximum line integral per unit area along the boundary of an infinitely small area at that point and the direction is perpendicular to this area. It is mathematically defined as

$$\text{Curl } \vec{A} = \vec{\nabla} \times \vec{A}$$

where $\vec{\nabla} = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$ and $\vec{A} = \hat{i} A_x + \hat{j} A_y + \hat{k} A_z$

$$\therefore \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \hat{i} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - \hat{j} \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) + \hat{k} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

Physical

(3)

- Stoke's Theorem: The line integral of a vector field $\vec{A}$ around the closed curve (C) is equal to the surface integral of the curl of $\vec{A}$ taken over any surface. It is mathematically expressed as $\int_C \vec{A} \cdot d\vec{l} = \int_S \text{curl } \vec{A} \cdot d\vec{S}$.

and it is used to transform surface integral into line integral.

- Gauss's Divergence Theorem: The normal surface integral of a vector function $\vec{A}$ over the boundary of a closed surface is equal to the volume integral of divergence of a vector field $\vec{A}$ over the volume V bounded by the surface S . It is mathematically expressed as

$$\int_V \text{div } \vec{A} dV = \int_S \vec{A} \cdot d\vec{S}$$

and is used to transform a volume integral into surface integral.

Some Useful Results:

(a) $\text{div grad } \phi = \vec{\nabla} \cdot \vec{\nabla} \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called Laplacian operator. It is a scalar function.

(b) $\text{curl grad } \phi = \vec{\nabla} \times \vec{\nabla} \phi = 0$ (c) $\text{div curl } \vec{A} = \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$

(d) $\text{curl curl } \vec{A} = \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \nabla \nabla \cdot \vec{A} - \nabla^2 \vec{A} = \text{grad div } \vec{A} - \nabla^2 \vec{A}$

Triple Vector Product:

$$\vec{A} \times \vec{B} \times \vec{C} = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - (\vec{\nabla} \cdot \vec{\nabla})\vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$$

• Evaluate curl grad ϕ

$$\begin{aligned} \vec{\nabla} \times \vec{\nabla} \phi &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \left(\hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z} \right) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} = \hat{i} \left[\frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial y} \right) \right] + \hat{j} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial x} \right) \right] \\ &\quad + \hat{k} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) \right] \end{aligned}$$

PHYSICAL SIGNIFICANCE OF CURL: In fact curl measures the twist of a vector.

1. When a rigid body is rotating about a fixed axis, then the curl of the linear velocity of a point on the body represents twice its angular velocity ($\text{curl } \vec{v} = 2\vec{\omega}$).
2. Curl of a magnetic field at nearby points of a current carrying conductor represents the current flowing per unit area to that point. Therefore, $\text{curl } \vec{B}$ is a magnetomotive force.

(4)

Electric Flux (Φ): The total number of electric lines of force passing normally through a surface in an electric field is called the electric flux linked with that surface. Unit of electric flux is Nm^2/C . The total flux through the entire surface is given by

$$\Phi_E = \int_S \vec{E} \cdot d\vec{S} = \int_S E \, dS \cos \theta$$

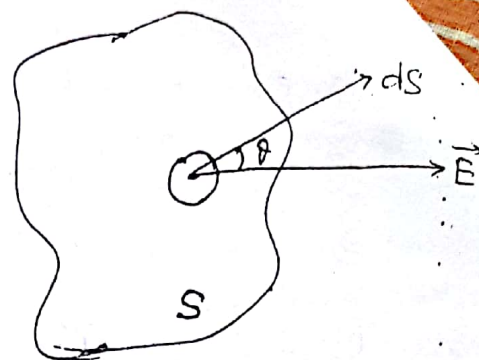


Fig. 1.4

Gauss's Law: Gauss's Law is the converse of Coulomb's law. (Gauss's Theorem) With the help of Coulomb's law we can calculate $\vec{E}$ for a given charge while Gauss's law enables to determine the charge provided $\vec{E}$ is known.

Gauss's law states that the net outward electric flux Φ_E through any closed surface is equal to $(1/\epsilon_0)$ the total charge (q) enclosed within the surface. Mathematically it can be expressed as -

$$\Phi_E = \int_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} q \quad \text{where } \epsilon_0 \text{ is the permittivity of free space.}$$

Proof: (a) When the charge is within the surface:

Let a charge q is placed at point O within a closed surface S , known as Gaussian surface as shown in Fig. 1.5.

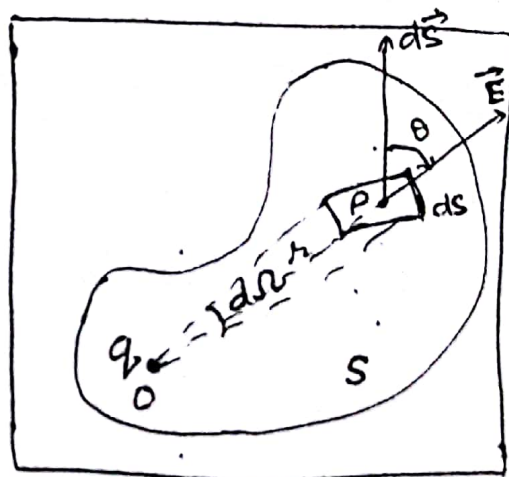


Fig. 1.5

Let us consider a small surface element of area dS around a point P on the surface. The area $d\vec{S}$ is directed normal to the surface. Let the distance of the point P from the point O is r . The intensity of electric field at point P due to charge q situated at point O is

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad \text{--- (1)}$$

If the θ is the angle b/w the electric field $\vec{E}$ and $d\vec{S}$, then the electric flux through the area $d\vec{S}$ is

$$d\phi_E = \vec{E} \cdot d\vec{S} = E dS \cos\theta \quad \text{--- (2)}$$

Substituting the value of E from eqn (1) into eqn (2)

we get

$$d\phi_E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} dS \cos\theta = \frac{q}{4\pi\epsilon_0} \frac{dS \cos\theta}{r^2}$$

$$\boxed{d\phi_E = \frac{q}{4\pi\epsilon_0} d\Omega} \quad \text{--- (3)}$$

where, $d\Omega$ is the solid angle subtended by the surface of area dS at the point O as shown in Fig. 1.5.

To find the total electric flux through the whole closed surface S , we divide the whole surface S in number of small surface elements and then add the electric flux through each of such element. Hence the total electric flux through the whole surface S is

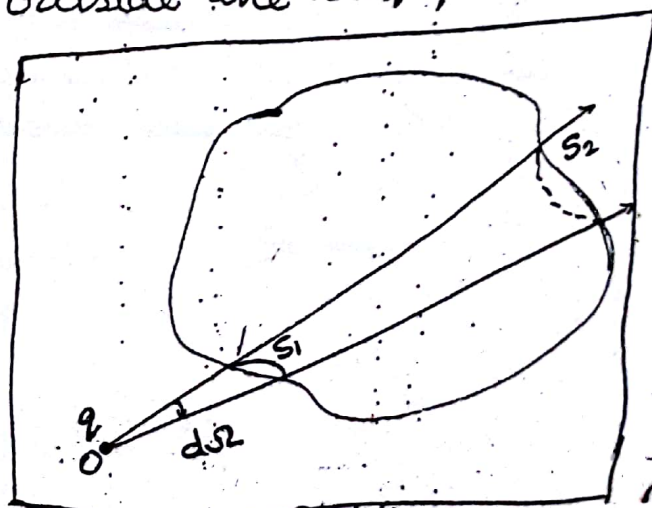
$$\phi_E = \int_S \frac{1}{4\pi\epsilon_0} q d\Omega = \frac{q}{4\pi\epsilon_0} \int_S d\Omega$$

$$\phi_E = \frac{1}{4\pi\epsilon_0} q [4\pi] = \frac{q}{\epsilon_0}$$

$$\phi_E = \int_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad \text{--- (4)}$$

Case II: When the charge lies outside the surface.

Let q be the charge situated at a point O outside the closed surface S , as shown in Fig. 1.6. Consider a cone subtending a solid angle $d\Omega$ at the point O , which intercepts



⑥

the given surface at S_1 and S_2 . The electric flux through the intercept S_1 is entering within the surface, so it is taken negative, while the electric flux through the intercepts S_2 is coming out of the surface, so it is taken positive.

From eq. (2), the electric flux entering through the intercept S_1 is $d\phi_1$ and is given by

$$d\phi_1 = -\frac{1}{4\pi\epsilon_0} q d\Omega$$

and the electric flux coming out through the intercept S_2 is $d\phi_2$ and is given by

$$d\phi_2 = +\frac{1}{4\pi\epsilon_0} q d\Omega$$

Therefore, the total electric flux through the surface enclosed by the intercepts S_1 and S_2 is

$$\begin{aligned} d\phi &= d\phi_1 + d\phi_2 \\ &= -\frac{1}{4\pi\epsilon_0} q d\Omega + \frac{1}{4\pi\epsilon_0} q d\Omega \\ d\phi &= 0 \end{aligned}$$

Applications of Gauss's Law (Gauss's Theorem):

Gauss's theorem can be applied to evaluate the electric field intensity at a point due to a charge or charge distribution. To find it, first we take a surface, called Gaussian surface, which is symmetrical for the given charge distribution and the point, where we want to evaluate the intensity of electric field lie on this surface. Then we find the electric flux through this surface with the help of Gauss's theorem and discussed as:

1. Electric field Intensity Due to a Point Charge:

Let there is a point charge q at a point O . We want to determine the electric field intensity at a point P , at a distance r from the point O (Fig. 1.7).

Consider a sphere of radius r with its centre at O .

The electric field at each point on the surface is directed normal to this surface. The flux through the surface is given by

$$\phi = \int_S \vec{E} \cdot d\vec{S} = \int_S E \, dS \cos \theta$$

$$\int E \, dS = E 4\pi r^2 \quad \text{--- ①}$$

According to Gauss's theorem

$$\phi = \int_S E \cdot dS = \frac{1}{\epsilon_0} q \quad \text{--- ②}$$

From eqs ① and ②, we get

$$E \times 4\pi r^2 = \frac{1}{\epsilon_0} q$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}}$$

which is Coulomb's law

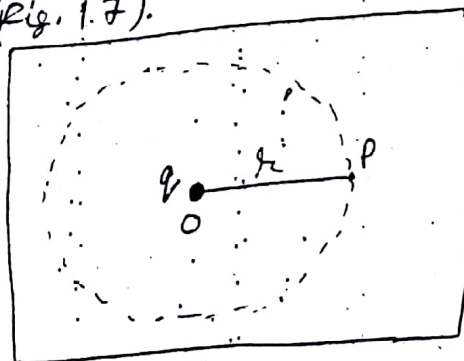


Fig. 1.7

2. Electric Field Intensity Due to Uniformly charged Conducting Sphere:

Case I: At a point (P) outside the charged sphere:

Let P is the point outside the sphere at a distance r from the centre O .

where, we want to find the electric field intensity. Construct a concentric Gaussian sphere of radius r as shown in Fig 1.8

The flux through the surface is given by

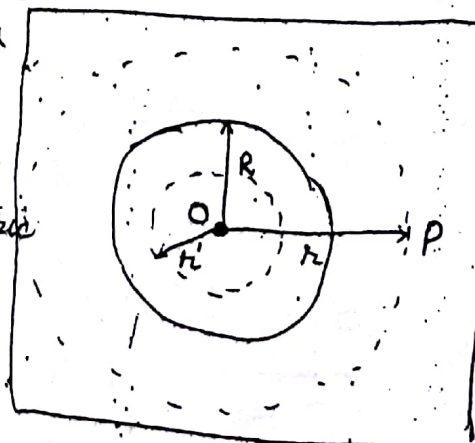


Fig. 1.8

⑧

$$\begin{aligned}\phi &= \int_S \vec{E} \cdot d\vec{S} = \int_S E dS \cos \theta \\ &= E \int dS = E \times 4\pi r^2\end{aligned} \quad \text{--- (1)}$$

According to Gauss's theorem,

$$\int_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} q \quad \text{--- (2)}$$

From eqs (1) and (2)

$$E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} q$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$\text{or } \boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}}$$

Case II: At a point on the surface: If the point P lies on the surface of sphere, then $r = R$.

$$\boxed{E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}}$$

Case-III: At a point inside the charged sphere: The point P is inside the sphere at a distance r' from the centre O (Fig. 1.8). Construct a concentric Gaussian sphere of radius r' ($r' < R$).

The outward flux through the surface is given by

$$\begin{aligned}\phi &= \int_S \vec{E} \cdot d\vec{S} = \int_S E dS \cos \theta = E \int dS \\ &= E \cdot 4\pi r'^2\end{aligned}$$

The total charge enclosed by the Gaussian surface = (Volume enclosed by it) $\times$ (Volume charge density)

$$q' = \frac{4}{3}\pi r'^3 \rho, \text{ where } \rho \text{ is the volume charge density.}$$

On applying the Gauss's theorem, we have

$$\int_S \vec{E} \cdot d\vec{S} = \frac{q'}{\epsilon_0}$$

$$E \cdot 4\pi r'^2 = \frac{q'}{\epsilon_0} = \frac{1}{\epsilon_0} \cdot \frac{4}{3}\pi r'^3 \rho = \frac{\rho}{3\epsilon_0} r'$$

$$\boxed{E = \frac{\rho}{3\epsilon_0} r'}$$

At the surface of sphere, $r' = R$

$$E = \frac{\rho}{3\epsilon_0} R$$

$$\rho = \frac{\text{charge}}{\text{Volume}} = \frac{q}{\frac{4}{3}\pi R^3}$$

$$E = \frac{1}{3\epsilon_0} \frac{q}{\frac{4}{3}\pi R^3} R = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

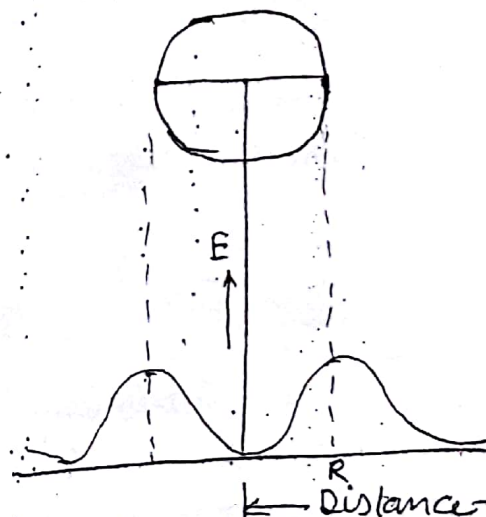


Fig. 1.9

Figure 1.9 shows the variation of electric field intensity with distance.

3. Electric Field Intensity Due to a Uniformly Charged Hollow Spherical Shell

Case I: The intensity of electric field at an external point:

Let q be the uniformly distributed over the spherical shell. Construct concentric Gaussian surface around this spherical shell of radius r as shown in Fig. 1.10. The flux

through the surface is given by —

$$\phi = \int_S \vec{E} \cdot d\vec{S} = E \int_S dS = E \cdot 4\pi r^2$$

According to Gauss's theorem,

$$\int \vec{E} \cdot d\vec{S} = q/\epsilon_0$$

where, q is the charge on the spherical shell.

From eqs. ① and ②, we get

$$E [4\pi r^2] = q/\epsilon_0$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

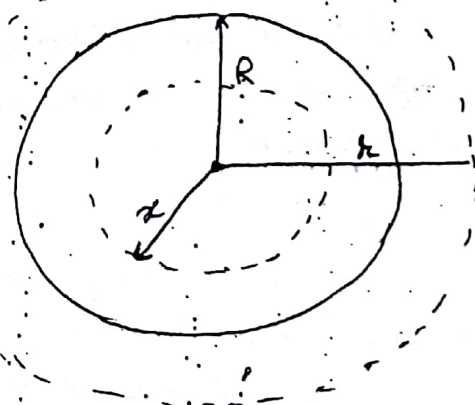


Fig. 1.10

(10)

Case II : At a Point on the Surface of the Charged Spherical Shell

If the point q lies on the surface of the spherical shell. Here $r = R$.

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$$

Case III : At a Point Inside the charged Spherical Shell :

The Gaussian surface inside the spherical shell encloses no charge, then the flux through this surface is equal to zero, i.e.,

$$E = 0$$

Thus, the intensity of electric field at each point inside a uniformly charged spherical shell is zero.

Figure 1.11 shows the variation of electric field E with the radial distance

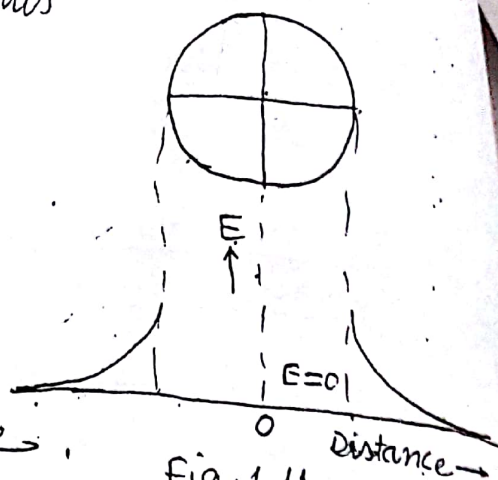


Fig. 1.11

4. Derivation of Coulomb's Law From Gauss's Theorem :

Let us imagine origin at the point charge and consider a spherical Gaussian surface of radius r . It is clear from symmetry that E must be radial and that its magnitude can depend only on the distance from the charge.

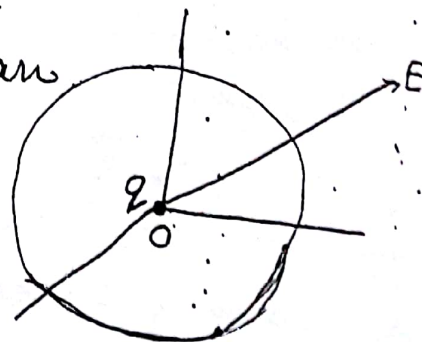


Fig. 1.12

The normal component $E \cdot \hat{n} = \vec{E} \cdot \hat{n}$ has the same value everywhere on the Gaussian spherical surface. The flux of E through this surface is this.

$$\phi = \oint E \cdot \hat{n} dS = \oint \vec{E} \cdot d\vec{S}$$

$$= E \oint dS = E \cdot 4\pi r^2$$

Since the total charge inside the surface is just the point charge q , Gauss's theorem gives -

$$E \cdot 4\pi r^2 = \frac{q}{\epsilon_0}$$

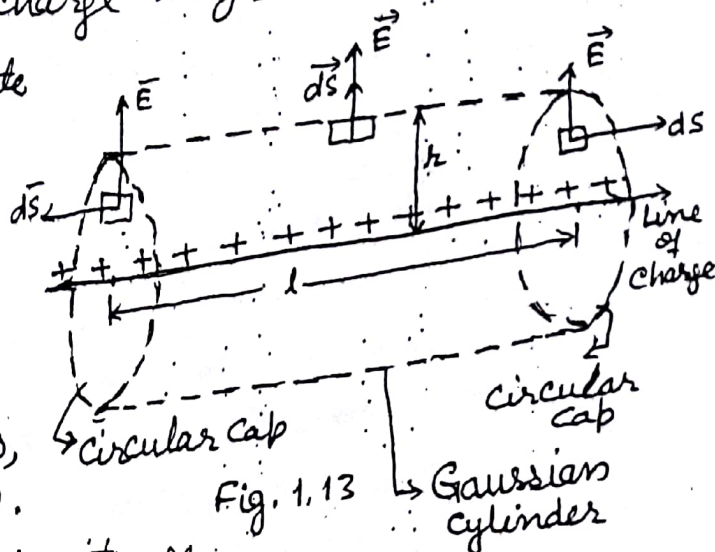
This is Coulomb law

$$\text{or } E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

5. Electric Field Due to An Infinite Line of Charge [Uniformly charged Long Wire]

An infinite line of charge may be a uniformly charged wire of infinite length.

Let us consider an infinite line of charge with a constant linear charge density λ (that is, charge per unit length).



In order to find the intensity of electric field at a distance r from the charged line, we draw a Gaussian surface around the line in the form of a circular cylinder of radius r and length l , closed at each end by plane parallel circular caps normal to the axis as shown in Fig. 1.13. All the points on the curved surface of this Gaussian cylinder are equivalent and at the same normal distance from the line charge. Due to symmetric distribution of charge the field everywhere on the curved surface of the Gaussian cylinder is equal in magnitude and radially outward in direction (for positive charge distribution). Therefore, for any patch of area dS taken on the cylinder (curved) surface, the field vector $\vec{E}$ and area vector $d\vec{S}$ both are along the same direction. Therefore,

$$d\phi = \vec{E} \cdot d\vec{S} = E dS \cos 0^\circ = E dS$$

Hence, the ^{electric} flux through the entire cylindrical surface (curved surface) of the Gaussian cylinder is

$$\phi = \oint \vec{E} \cdot d\vec{S} = \oint E dS = E \oint dS$$

But $\oint dS = 2\pi r l$, the area of the cylindrical surface

$$\therefore \phi = E \times (2\pi r l) \quad \text{--- (1)}$$

②

The electric flux through the circular flat caps of the Gaussian cylinder is zero because the electric flux at any point on either of the caps is perpendicular to the line charge or to the area vectors on these caps.

Hence, $\vec{E}$ and $d\vec{S}$ are at an angle 90° with each other (Fig. 1.13) so that $\oint \vec{E} \cdot d\vec{S} = 0$ on both the caps.

Thus, the total flux through the closed Gaussian cylinder is therefore,

$$\phi = \oint \vec{E} \cdot d\vec{S} = E(2\pi r l) \quad \text{--- ②}$$

The total charge enclosed by the Gaussian surface = λl

According to Gauss's law

$$\phi = \oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad \text{or} \quad \boxed{\phi = \frac{\lambda l}{\epsilon_0}} \quad \text{--- ③}$$

From eqs ② and ③, we have

$$E \cdot 2\pi r l = \frac{\lambda l}{\epsilon_0} \quad \text{or} \quad \boxed{E = \frac{\lambda}{2\pi \epsilon_0 r}} \quad \text{--- ④}$$

It is apparent from equation ④ that the field is inversely proportional to r . It is radially outward in direction if the charge is +ve and radially inward if the charge is -ve.

6. Electric Field due to an Infinite Plane of Uniform Charge Density (OR Charge Insulating Sheet)

Let us consider a portion of a, thin, non-conducting, infinite plane sheet of charge with constant surface charge density (σ)

(Charge per unit area).

Suppose we want to find the intensity of electric field $\vec{E}$ at a point P near the sheet, distance r in front of the sheet (Fig. 1.14). To evaluate the field

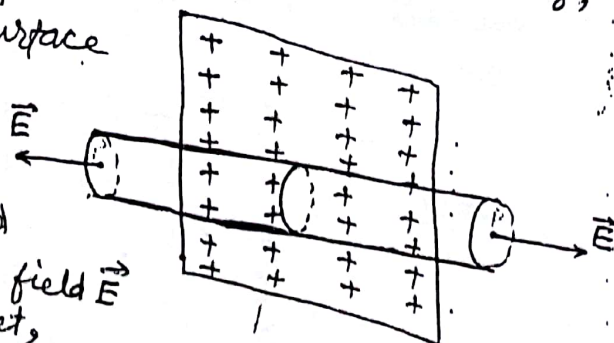


Fig. 1.14

at P and P' are equidistant from the sheet. Now we draw a small closed Gaussian cylinder with its circular ends parallel to the sheet and passed through the points P and P'. Suppose the flat ends at P and P' have equal area dS . The cylinder together with flat ends form a closed surface, such that the Gauss's law can be applied.

By symmetry, the magnitude of the electric field $\vec{E}$ at all the points on either side end caps is same and along the outward normal (for +vely charged sheet). Therefore, on either side end caps the field vector $\vec{E}$ and area vector $d\vec{S}$ are parallel to each other, that is, the angle b/w them is zero. Hence, the electric flux through each cap is

$$d\phi = \vec{E} \cdot d\vec{S} = E dS \cos 0^\circ = E dS$$

At the points on the curved surface the field vector $\vec{E}$ and area vector $d\vec{S}$ makes an angle 90° with each other and hence $d\phi = \vec{E} \cdot d\vec{S} = E dS \cos 90^\circ = 0$, therefore, cylindrical surface does not contribute to the flux. Hence, the total flux through the closed surface is

$$\phi = \oint \vec{E} \cdot d\vec{S} = \int E \cdot dS + 0$$

$$\phi = 2ES$$

According to Gauss's law

$$\phi = 2ES = \frac{q}{\epsilon_0} \quad \text{or} \quad E = \frac{q}{2\epsilon_0 S} = \frac{\sigma S}{2\epsilon_0 S}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

From this expression it is apparent that the field is uniform and does not depend on the distance from the plane sheet of charge. This conclusion holds good at all points near the sheet.

(14)

Differential Form of Gauss's Theorem

When charge is distributed over a volume such that ρ is the volume charge density, then the charge enclosed by surface enclosing the volume is given by

$$q = \int_V \rho \, dv \quad \text{--- (1)}$$

By Gauss's law

$$\int_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad \text{--- (2)}$$

Put the value of q in eq (2) from eq. (1), we get

$$\int_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \int_V \rho \, dv \quad \text{--- (3)}$$

According to Gauss's divergence theorem, we can convert the surface integral to volume integral as -

$$\int_S \vec{E} \cdot d\vec{S} = \int_V \text{div } \vec{E} \, dv \quad \text{--- (4)}$$

By using eq. (3) and (4), we get

$$\int \text{div } \vec{E} \, dv = \frac{1}{\epsilon_0} \int \rho \, dv$$

$$\int (\text{div } \vec{E} - \rho/\epsilon_0) \, dv = 0$$

The volume is arbitrary, hence it is true for any volume, so

$$\text{div } \vec{E} - \rho/\epsilon_0 = 0 \quad \text{or} \quad \text{div } \vec{E} = \rho/\epsilon_0$$

$$\text{or} \quad \epsilon_0 \text{div } \vec{E} = \rho \quad \text{or} \quad \text{div}(\epsilon_0 \vec{E}) = \rho$$

$$\text{or} \quad \boxed{\text{div } \vec{D} = \rho} \quad \text{--- (5)}$$

where, $\vec{D}$ is electric flux density.

Equation (5) is the differential form of Gauss's Theorem.

Ampere's Circuital Law:

Ampere's circuital law in magnetostatics is analogous to the Gauss's law in electrostatics. This law says that the line integral of magnetic field $\vec{B}$ around any closed loop is equal to μ_0 times the net current I flowing through the area enclosed by the loop i.e.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$$

Here μ_0 is the permeability of the free space.

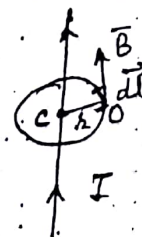


Fig. 1.4

Proof: Consider a long straight conductor carrying a current I . By Biot-Savart law, the magnitude of the magnetic field at a point O, at a distance r from the conductor, is given by

$$B = \frac{\mu_0}{4\pi} \frac{2I}{r} \quad \text{--- (1)}$$

Let us draw a circle with a radius r taking C as centre around the current carrying conductor as shown in Fig. 1.4. The intensity of magnetic field $\vec{B}$ will be the same in magnitude at all points on this circle. Again we consider a small element on the circle of length dl at the point O. From the figure 1.4, it is clear that $d\vec{l}$ and $\vec{B}$ are in the same direction

$$\text{Therefore } \oint_C \vec{B} \cdot d\vec{l} = \oint B dl \cos \theta$$

$$= B \oint dl$$

$$\{\because \cos \theta = \cos 0^\circ = 1\}$$

$$\text{or } \oint_C B \cdot dl = \frac{\mu_0}{4\pi} \cdot \frac{2I}{r} \cdot 2\pi r = \mu_0 I$$

$$\left\{ \begin{array}{l} \because B = \frac{\mu_0}{4\pi} \cdot \frac{2I}{r} \\ \text{By Biot-Savart law} \end{array} \right\}$$

$$\boxed{\oint_C B \cdot dl = \mu_0 I}$$

This is Ampere's law

Stoke's Theorem: Relates line integral to the surface integral. Line integral of $\vec{A}$ around a closed curve is equal to Surface integral of the curl of a vector field $\vec{A}$ over any surface S .

$$\boxed{\oint_C \vec{A} \cdot d\vec{l} = \int_S \text{Curl } \vec{A} \cdot d\vec{s}}$$

Gauss Divergence Theorem: Surface integral of vector field $\vec{A}$ is equal to divergence of $\vec{A}$ over the volume V bounded by the surface S .

$$\int_S \vec{A} \cdot d\vec{s} = \int_V \text{div } \vec{A} \cdot dV$$

①⑥ Equation of Continuity:

It states that total current flowing out of some volume must be equal to the rate of decrease of charge within the volume, assuming that charge can not be created or destroyed.

The rate of flow of charge (due to -ve. ly charged electrons) constitute electric current, i.e.

$$I = - \frac{dq}{dt} = - \frac{d}{dt} \int_V \rho dV \quad \text{--- ①}$$

The current (I) may also be expressed in terms of current density (J) as

$$I = \int_S \vec{J} \cdot d\vec{S} \quad \text{--- ②} \quad \left(\because J = \frac{I}{A} \right)$$

From eq. ① and ②, we have

$$\int_S \vec{J} \cdot d\vec{S} = - \frac{d}{dt} \int_V \rho dV \quad \text{--- ③}$$

From Gauss's divergence theorem, we know that

$$\int_S \vec{J} \cdot d\vec{S} = \int_V \text{div } \vec{J} dV \quad \text{--- ④}$$

By using eq. ③ and ④, we have

$$\int_V \text{div } \vec{J} dV = - \int_V \frac{\partial \rho}{\partial t} dV$$

$$\text{or } \int_V \left(\text{div } \vec{J} + \frac{\partial \rho}{\partial t} \right) dV = 0 \quad \text{--- ⑤}$$

The volume is arbitrary so it is true for any arbitrary volume so

$$\boxed{\text{div } \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

This is equation of continuity.

In case of stationary current, charge density remains constant, i.e. $\frac{\partial \rho}{\partial t} = 0$, so that

$$\boxed{\text{div } \vec{J} = 0}$$

It means that there is no net outward flux of the current density, J.

Maxwell's Equations (Differential forms)

(17)

$$1. \operatorname{div} \vec{E} = \rho / \epsilon_0$$

$$2. \operatorname{div} \vec{B} = 0$$

$$3. \operatorname{curl} \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$4. \operatorname{curl} \vec{B} = \mu_0 \vec{J} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{D} = \rho$$

$$\vec{\nabla} \cdot \vec{H} = 0$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

When the charges are in motion, the electric and magnetic fields are associated with this motion which will have variations in both the space and time. These electric and magnetic fields are inter-related. This phenomenon is called 'electromagnetism' which is summarised by the set of equations, known as Maxwell's equations.

The Maxwell's equations are nothing but are the representation of the basic laws of electromagnetism.

In the differential form, four Maxwell's equations are given below:

$$1. \vec{\nabla} \cdot \vec{D} = \rho$$

$$2. \vec{\nabla} \cdot \vec{B} = 0$$

$$3. \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$4. \vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \frac{\partial \vec{D}}{\partial t} \mu_0$$

Maxwell's Equations In Integral Form:

There are situations where the integral form of Maxwell's equations is useful.

$$1. \oint_S \vec{D} \cdot d\vec{S} = q = \int_V \rho \, dV$$

$$2. \oint_S \vec{B} \cdot d\vec{S} = 0$$

$$3. \oint_C \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{S}$$

$$4. \oint_C \vec{H} \cdot d\vec{l} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{S}$$

Maxwell's Equations In Isotropic Dielectric Medium:

In an isotropic dielectric medium, the current density $\vec{J}$ and the volume charge density ρ are zero. Further, the displacement vector $\vec{D}$ and the magnetic field $\vec{B}$ are defined as $\vec{D} = \epsilon \vec{E}$ and $\vec{B} = \mu \vec{H}$. In fact $\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon \vec{E}$ and $\vec{B} = \mu_0 (\vec{M} + \vec{H})$ for the isotropic linear dielectric medium. Here, the vectors, $\vec{P}$ and $\vec{M}$ gives the measure of polarisation and magnetisation of the medium respectively. However for the

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dielectric medium, ϵ_0 and μ_0 of the free space have been simply replaced with ϵ and μ . Hence for dielectric medium

$$J = 0 \quad (= \sigma E) = 0 \quad (\text{or } \sigma = 0), \quad \rho = 0$$

$$\begin{array}{ll} 1. \quad \vec{\nabla} \cdot \vec{E} = 0 & 3. \quad \vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \\ 2. \quad \vec{\nabla} \cdot \vec{H} = 0 & 4. \quad \vec{\nabla} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t} \end{array}$$

Maxwell's Equations In Conducting Medium:

We consider a conducting linear and isotropic medium whose permeability is μ , permittivity is ϵ and conductivity is σ . Further, we assume that there is no free charge available, i.e. $\rho = 0$ (inside the conductor because all the charge immediately spread on the surface i.e. inside charge will be zero). Under this situation, we can write the Maxwell's equations as

$$\begin{array}{ll} 1. \quad \vec{\nabla} \cdot \vec{E} = 0 & 3. \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t} \\ 2. \quad \vec{\nabla} \cdot \vec{B} = 0 & 4. \quad \vec{\nabla} \times \vec{H} = J + \epsilon \frac{\partial \vec{E}}{\partial t} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \end{array}$$

Derivations of Maxwell's Equations (Integral Forms)

(19)

(I) Derivation of Maxwell's 1st Equation: Let us consider a surface bounding a volume V in a dielectric medium, which is kept in the electric field $\vec{E}$. The application of external field $\vec{E}$ polarises the dielectric medium and charges are induced, called bound charges or charges due to polarisation. The total charge density at a point in a small volume element dV would then be $(\rho + \rho_p)$, where ρ_p is the polarisation charge density, given by $\rho_p = -\text{div } P$ and ρ is the free charge density at that point in a small volume element dV . Thus the total charge density at that point will be $(\rho - \text{div } P)$. Then by Gauss's theorem

$$\int_S \vec{E} \cdot d\vec{S} = \int_V \text{div } \vec{E} dV = \frac{1}{\epsilon_0} \int_V (\rho - \text{div } P) dV$$

$$\text{or } \epsilon_0 \int_V \text{div } \vec{E} dV = \int_V (\rho - \text{div } P) dV$$

$$\text{or } \int_V \text{div} (\epsilon_0 \vec{E} + P) = \int_V \rho dV$$

The quantity $(\epsilon_0 \vec{E} + P)$ is denoted by the quantity $\vec{D}$, called the electric displacement. Therefore

$$\int_V \text{div } \vec{D} dV = \int_V \rho dV$$

Since the equation is true for all the arbitrary volumes, the integrands in this must be equal to

$$\boxed{\text{div } \vec{D} = \rho}$$

$$\text{or } \boxed{\vec{\nabla} \cdot \vec{D} = \rho}$$

This is Maxwell's 1st Equation.

(II) Derivation of Maxwell's 2nd Equation: Since the magnetic line of force are either closed or go off to infinity, the number of magnetic line of force entering any arbitrary surface is exactly the same as leaving it. It means the flux of magnetic induction $\vec{B}$ across any closed surface is always zero, i.e. $\int_S \vec{B} \cdot d\vec{S} = 0$

Transforming the surface integral to the volume integral using Gauss's divergence theorem, we have

$$\int_S \vec{B} \cdot d\vec{S} = \int_V \text{div } \vec{B} dV = 0$$

for any arbitrary volume
 $\boxed{\text{div } \vec{B} = 0}$ This is Maxwell's 2nd Equation.

20 (III) Derivation of Maxwell's IIIrd Equation:

According to Faraday's law, the e.m.f. induced in the closed loop is given by

$$E_{emf} = -\frac{\partial \Phi_B}{\partial t} = -\frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{S} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

Here the flux $\Phi_B = \int_S \vec{B} \cdot d\vec{S}$, where S is any closed surface having the loop as boundary. The e.m.f. (E_{emf}) can also be found by calculating the work done in carrying a unit charge completely around the loop. Thus,

$$E_{emf} = \oint_C \vec{E} \cdot d\vec{l}$$

Here $\vec{E}$ is the intensity of the field associated with the induced e.m.f. on equating the above two equations, we get

$$\oint_C \vec{E} \cdot d\vec{l} = -\oint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad \text{--- (1)}$$

According to Stoke's theorem, the line integral can be transformed into surface integral, i.e.

$$\oint_C \vec{E} \cdot d\vec{l} = \int_S (\nabla \times \vec{E}) \cdot d\vec{S} \quad \text{--- (2)}$$

From eqs (1) and (2), then

$$\int_S \nabla \times \vec{E} \cdot d\vec{S} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} \quad \text{--- (3)}$$

This equation must be true for any surface. So the two vectors in the integrands must be equal at every point, i.e.,

$$\boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}} \quad \text{This is Maxwell's IIIrd equation.}$$

(IV) Derivation of Maxwell's IVth Equation: According to Ampere's law, the work done in carrying the unit magnetic pole once around a closed arbitrary path linked with the current is expressed by

$$\oint_C \vec{H} \cdot d\vec{l} = I = \int_S \vec{J} \cdot d\vec{S} \quad \text{--- (1)}$$

As per Stoke's theorem

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \nabla \times \vec{H} \cdot d\vec{S} \quad \text{--- (2)}$$

$$\text{Therefore } \int_S \nabla \times \vec{H} \cdot d\vec{S} = \int_S \vec{J} \cdot d\vec{S} \quad \text{--- (3)}$$

$$\text{This gives } \boxed{\nabla \times \vec{H} = \vec{J}} \quad \text{or } \boxed{\text{Curl } \vec{H} = \vec{J}} \quad \text{--- (4)}$$

The above relation is derived on the basis of Ampere's law, which holds good only for the steady current however, for changing electric fields, the current density should be modified. The difficulty with the above equation is that,

if we take divergence of the above equation, then

$$\text{div}(\vec{\nabla} \times \vec{H}) = \text{div } \vec{J} \quad \left\{ \text{Since the div of curl of vector} = 0 \right\}$$

$$0 = \text{div } \vec{J}$$

or $\boxed{\text{div } \vec{J} = 0}$ — (5)

which conflicts with the continuity equation, as

$$\boxed{\text{div } \vec{J} = -\frac{\partial \rho}{\partial t}} \text{ — (6) } \{ \text{continuity eqn} \}$$

Therefore, Maxwell realised that the definition of total current density is incomplete and suggested to add another density $\vec{J}'$.

Therefore $\text{Curl } \vec{H} = \vec{J} + \vec{J}'$ — (7)

Now taking divergence of the above equation, we get

$$\text{div}(\text{Curl } \vec{H}) = \text{div } \vec{J} + \text{div } \vec{J}'$$
$$0 = \text{div } \vec{J} + \text{div } \vec{J}'$$

$$\text{div } \vec{J}' = -\text{div } \vec{J} = \frac{\partial \rho}{\partial t} \text{ — (8)}$$

Since $\rho = \vec{\nabla} \cdot \vec{D}$ — (9)

$$\text{So } \text{div } \vec{J}' = \frac{\partial}{\partial t}(\vec{\nabla} \cdot \vec{D}) = \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t} \text{ — (10)}$$

$$\text{Hence } \vec{J}' = \frac{\partial \vec{D}}{\partial t} \text{ — (11)}$$

So, by using eqn (7) and (11), then, we get

$$\text{Curl } \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

or $\boxed{\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}}$ — (12) This is the Maxwell's IVth equation.

The last term of R.H.S. of equation (12) is called the Maxwell's correction and is known as displacement current density. The above equation is called modified Ampere's law for unsteady or changing current which is responsible for electromagnetic fields.

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Significance of Maxwell's Equations:

Maxwell's Ist equation represents the Gauss's law in electrostatics, which states that the electric flux over Gaussian surface is $1/\epsilon_0$ the total charge enclosed within the surface.

Maxwell's IInd equation represents the Gauss's in magnetostatics, which states that the total magnetic flux through any closed surface is zero.

Maxwell's IIIrd equation is the Faraday's law of electromagnetic induction and signifies that an electric field is produced by changing magnetic flux.

Maxwell's IVth equation is the modified form of Ampere's circuital law, which states that magnetomotive force around a closed path is equal to sum of conduction current and displacement current.

Poynting Vector and Poynting Theorem

The electromagnetic wave carry energy when they propagate and there is an energy density associated with both electric and magnetic fields.

The amount of energy flowing through unit area, perpendicular to the direction of energy propagation per unit time, i.e. the rate of energy transport per unit area is called poynting vector. It is also termed as instantaneous energy flux density and is represented by $\vec{S}$. Mathematically it is represented by

$$\boxed{\vec{S} = \vec{E} \times \vec{H}}$$

where $\vec{E}$ and $\vec{H}$ represent the instantaneous values of electric and magnetic fields vectors. This is clear that the rate of energy transport $\vec{S}$ is perpendicular to both $\vec{E}$ and $\vec{H}$ and is in the direction of propagation of the wave, as $\vec{E} \times \vec{H}$ is in the direction of $\vec{k}$. Since the poynting vector represents the rate of energy transport per unit area. Its units are W/m^2 .

Poynting Vector: E.M. waves transport energy as they propagate from one region to another. This transfer of energy can be described in terms of energy transferred per unit time per unit cross-sectional area or power per unit area for an area perpendicular to the direction of propagation of the wave.

Derivation of Poynting Theorem: We can calculate the energy density carried by electromagnetic waves with the help of Maxwell's equations given below.

$$\begin{aligned} \text{div } \vec{D} &= 0 & \text{--- (i)} & \quad \text{curl } \vec{E} = -\frac{\partial \vec{B}}{\partial t} & \text{--- (iii)} \\ \text{div } \vec{B} &= 0 & \text{--- (ii)} & \quad \text{curl } \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} & \text{--- (iv)} \end{aligned}$$

Take dot product (or scalar product) of eqn. (iii) and eqn. (iv) with $\vec{H}$ and $\vec{E}$ respectively, i.e.

$$\begin{aligned} \vec{H} \cdot \text{curl } \vec{E} &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} & \text{--- (v)} \\ \text{and } \vec{E} \cdot \text{curl } \vec{H} &= \vec{E} \cdot \vec{J} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} & \text{--- (vi)} \end{aligned}$$

Subtract eqn. (vi) from eqn. (v), i.e.

$$\begin{aligned} \vec{H} \cdot \text{curl } \vec{E} - \vec{E} \cdot \text{curl } \vec{H} &= -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \vec{J} - \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \\ &= -\left[\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right] - \vec{E} \cdot \vec{J} \\ \text{or } \text{div}(\vec{E} \times \vec{H}) &= -\left[\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} + \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} \right] - \vec{E} \cdot \vec{J} \\ & \quad \left[\because \text{div}(\vec{A} \times \vec{B}) = \vec{B} \cdot \text{curl } \vec{A} - \vec{A} \cdot \text{curl } \vec{B} \right] \end{aligned}$$

Using the relations $\vec{B} = \mu \vec{H}$ and $\vec{D} = \epsilon \vec{E}$, we can get

$$\vec{E} \cdot \frac{\partial \vec{D}}{\partial t} = \vec{E} \cdot \frac{\partial (\epsilon \vec{E})}{\partial t} = \frac{1}{2} \epsilon \frac{\partial (\vec{E} \cdot \vec{E})}{\partial t} = \frac{1}{2} \epsilon \frac{\partial (E^2)}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \vec{E} \cdot \vec{D} \right)$$

Similarly

$$\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} = \vec{H} \cdot \frac{\partial (\mu \vec{H})}{\partial t} = \frac{1}{2} \mu \frac{\partial (H^2)}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \vec{H} \cdot \vec{B} \right)$$

Now eqn. (vii) can be written as

$$\begin{aligned} \text{div}(\vec{E} \times \vec{H}) &= -\frac{\partial}{\partial t} \left[\frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D}) \right] - \vec{E} \cdot \vec{J} \\ \text{or } \vec{E} \cdot \vec{J} &= -\frac{\partial}{\partial t} \left[\frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D}) \right] - \text{div}(\vec{E} \times \vec{H}) & \text{--- (viii)} \end{aligned}$$

Integrating eqn. (viii) over a volume V enclosed by a surface S , we get

$$\begin{aligned} \int_V \vec{E} \cdot \vec{J} \, dV &= -\int_V \left[\frac{\partial}{\partial t} \left\{ \frac{1}{2} (\vec{H} \cdot \vec{B} + \vec{E} \cdot \vec{D}) \right\} \right] dV - \int_V \text{div}(\vec{E} \times \vec{H}) \, dV \\ \text{or } \int_V \vec{E} \cdot \vec{J} \, dV &= -\int_V \left[\frac{\partial}{\partial t} \left(\frac{1}{2} \mu H^2 + \frac{1}{2} \epsilon E^2 \right) \right] dV - \int_S (\vec{E} \times \vec{H}) \cdot d\vec{S} & \text{--- (ix)} \end{aligned}$$

$\because \vec{B} = \mu \vec{H}$, $\vec{D} = \epsilon \vec{E}$ and $\int_V \text{div}(\vec{E} \times \vec{H}) \, dV = \int_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$
Equation (ix) can be written as

$$\boxed{\int_V \vec{E} \cdot \vec{J} \, dV = -\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \mu H^2 + \frac{1}{2} \epsilon E^2 \right] dV - \int_S (\vec{E} \times \vec{H}) \cdot d\vec{S}} \quad \text{--- (x)}$$

$$(26) \vec{\nabla} \cdot \vec{E} = \int \hat{i} \cdot \hat{a} e^{i(x\hat{x} + y\hat{y})}$$

(24)

Interpretation:

- (i) $\int_V (\vec{E} \cdot \vec{J}) dV$: This term represents the rate of energy transferred into the electromagnetic field through the motion of charges in the volume V , i.e. total power dissipated in a volume V .
- (ii) $\frac{\partial}{\partial t} \int_V \left[\frac{1}{2} \mu H^2 + \frac{1}{2} \epsilon E^2 \right] dV$: The term $\frac{1}{2} \mu H^2$ and $\frac{1}{2} \epsilon E^2$ represent the energy stored in electric and magnetic fields respectively and their sum will be equal to the total energy stored in electromagnetic fields. Therefore, this total expression represents the rate of decrease of energy stored in volume V due to electric and magnetic fields.
- (iii) $\int_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$: This term represents the amount of electromagnetic energy crossing the closed surface per second or the rate of flow of outward energy through the surface S enclosing volume V . The vector $(\vec{E} \times \vec{H})$ is known as the Poynting vector $\vec{S}$ or $\vec{S} = (\vec{E} \times \vec{H})$.

Equation (x) is known as Poynting theorem.

$$\vec{A} \times \vec{B} \times \vec{C} = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$$

Propagation of Electromagnetic wave in Free space (25)

Maxwell's equations for free space are given as follows

$$\text{div } \vec{E} = 0 \quad \text{--- (i)}$$

$$\text{div } \vec{H} = 0 \quad \text{--- (ii)}$$

$$\text{curl } \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{--- (iii)}$$

$$\text{curl } \vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{--- (iv)}$$

Taking curl of eqn (iii), we get

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\mu_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H})$$

$$\text{or } \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \frac{\partial}{\partial t} (\epsilon_0 \frac{\partial \vec{E}}{\partial t}) \quad \text{[Using eqn (iv)]}$$

Now from eqn (i) $\vec{\nabla} \cdot \vec{E} = 0$, Hence

$$\nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad \text{--- (v)}$$

This is the wave equation governing the field $\vec{E}$. In view of the dimensions of $\mu_0 \epsilon_0$ as of velocity (say v), we can write this as

$$\boxed{\nabla^2 \vec{E} - \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0} \quad \text{--- (v)}$$

Similarly taking curl of eqn (iv) gives the wave equation for the field $\vec{H}$ as

$$\nabla^2 \vec{H} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2} = 0$$

$$\text{or } \boxed{\nabla^2 \vec{H} - \frac{1}{v^2} \frac{\partial^2 \vec{H}}{\partial t^2} = 0} \quad \text{--- (vi)}$$

The plane wave solutions of eqns. (v) and (vi) may be written as

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{--- (vii)} \quad \text{and} \quad \vec{H}(\vec{r}, t) = \vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{--- (viii)}$$

where ω is the angular frequency of the variation of the fields $\vec{E}$ and $\vec{H}$ and $\vec{k}$ is the wave vector which tells the direction of propagation of the wave. The ratio ω/k gives the phase velocity of the wave. The symbol i (iota) = $\sqrt{-1}$.

$$\text{Now, } \vec{\nabla} = \left\{ \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right\}, \quad \vec{E}_0 = (E_{0x} \hat{i} + E_{0y} \hat{j} + E_{0z} \hat{k}),$$

$$\vec{k} = (k_x \hat{i} + k_y \hat{j} + k_z \hat{k}) \quad \text{and} \quad \vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\Rightarrow \vec{k} \cdot \vec{r} = k_x x + k_y y + k_z z$$

The solutions (vii) and (viii) must satisfy not only the wave eqns. (v) and (vi), but also all the Maxwell's eqns. We now find $\vec{\nabla} \cdot \vec{E}$

$$\vec{\nabla} \cdot \vec{E} = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \cdot \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} = 0$$

$$\begin{aligned} \textcircled{26} \quad \vec{\nabla} \cdot \vec{E} &= \left\{ \hat{i} \frac{\partial}{\partial x} e^{i(k_x x + y k_y + z k_z) - \omega t} + \hat{j} \frac{\partial}{\partial y} e^{i(k_x x + y k_y + z k_z) - \omega t} + \hat{k} \frac{\partial}{\partial z} e^{i(k_x x + y k_y + z k_z) - \omega t} \right\} \cdot [\hat{i} E_{0x} + \hat{j} E_{0y} + \hat{k} E_{0z}] \\ &= [\hat{i} i k_x + \hat{j} i k_y + \hat{k} i k_z] e^{i(\vec{k} \cdot \vec{r} - \omega t)} \cdot [\hat{i} E_{0x} + \hat{j} E_{0y} + \hat{k} E_{0z}] \\ &= i [\hat{i} k_x + \hat{j} k_y + \hat{k} k_z] \cdot (\hat{i} E_{0x} + \hat{j} E_{0y} + \hat{k} E_{0z}) \cdot e^{i(\vec{k} \cdot \vec{r} - \omega t)} \end{aligned}$$

$$\vec{\nabla} \cdot \vec{E} = i \vec{k} \cdot \vec{E} e^{i(\vec{k} \cdot \vec{r} - \omega t)} = i \vec{k} \cdot \vec{E} \quad \text{--- (ix)}$$

Similarly, it can be shown that

$$\vec{\nabla} \cdot \vec{H} = i \vec{k} \cdot \vec{H} \quad \text{--- (x)}$$

Thus, from eqns (ix) and (x) we find the operator $\vec{\nabla}$ transforms into another operator $i \vec{k}$

$$\vec{\nabla} \Rightarrow i \vec{k} \quad \text{--- (xi)}$$

where $i = \sqrt{-1}$ and k is the propagation vector of e.m. wave

The Maxwell's eqn $\vec{\nabla} \cdot \vec{E} = 0$ and $\vec{\nabla} \cdot \vec{H} = 0$ demand that

$$\vec{k} \cdot \vec{E} = 0 \quad \text{and} \quad \vec{k} \cdot \vec{H} = 0$$

$$\therefore \vec{\nabla} \cdot \vec{E} = 0 \quad \text{or} \quad i \vec{k} \cdot \vec{E} = 0 \quad \text{or} \quad \vec{k} \cdot \vec{E} = 0$$

From the above results it is clear that both $\vec{E}$ and $\vec{H}$ vectors are perpendicular to the direction of propagation vector $\vec{k}$ (direction of propagation of e.m. waves).

Now, ~~considering~~ ~~differentiating~~ of eqns. (iii) and (iv), then we get -

$$\text{Curl } \vec{E} = \vec{\nabla} \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{--- (xii)} \quad \text{and}$$

$$\text{Curl } \vec{H} = \vec{\nabla} \times \vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{--- (xii)}$$

$$\text{Now } \vec{H}(r, t) = \vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{and} \quad \vec{E}(r, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

~~differentiating~~ ~~of above eqn w.r. to~~ t , then {From eqns (iii) & (iv)}

$$\frac{\partial \vec{H}}{\partial t} = \frac{\partial}{\partial t} [\vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}] = \vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \times (-i\omega)$$

$$\frac{\partial \vec{H}}{\partial t} = -i\omega \vec{H} \quad \therefore \text{Thus } \frac{\partial}{\partial t} \Rightarrow -i\omega$$

$$\text{Similarly } \frac{\partial \vec{E}}{\partial t} = \frac{\partial}{\partial t} [\vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}] = -i\omega \vec{E}$$

$$\text{Thus } \frac{\partial}{\partial t} \Rightarrow -i\omega$$

Using these results, hence eqs. (xi) and (xii) become

$$i\vec{k} \times \vec{E} = -\mu_0 (-i\omega \vec{H}) \text{ or } \vec{k} \times \vec{E} = \mu_0 \omega \vec{H} \quad \text{--- (xiii)}$$

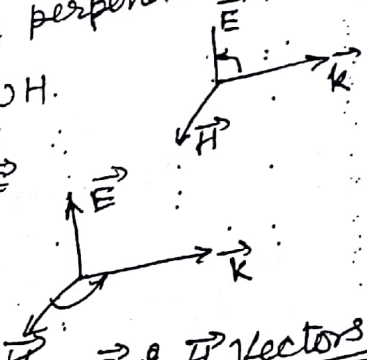
$$\text{and } i\vec{k} \times \vec{H} = \epsilon_0 (-i\omega \vec{E}) \text{ or } \vec{k} \times \vec{H} = -\epsilon_0 \omega \vec{E} \quad \text{--- (xiv)}$$

From eq. (xiii), it is clear that $\vec{H}$ is $\perp \vec{k}$ & $\vec{H} \perp \vec{E}$.

Hence Vectors $\vec{E}$, $\vec{H}$ and $\vec{k}$ are mutually perpendicular.

Equation (xiii) implies that $\vec{k} \times \vec{E} = \mu_0 \omega \vec{H}$.

Equation (xiv) implies that $\vec{H} \times \vec{k} = \epsilon_0 \omega \vec{E}$.



Wave Impedance of Free space and Phase of $\vec{E}$ & $\vec{H}$ Vectors:

From equation (xiii)

$$\vec{H} = \frac{\vec{k} \times \vec{E}}{\mu_0 \omega} = \frac{k}{\mu_0 \omega} (\hat{n} \times \vec{E}) \quad \text{--- (xv) } (\because \vec{k} = \hat{n} k)$$

$$\text{Also } k = \frac{2\pi}{\lambda} \times \frac{\nu}{\nu} = \frac{2\pi \nu}{\lambda} = \frac{\omega}{c} \text{ or } \boxed{\frac{k}{\omega} = \frac{1}{c}} \quad \text{--- (xvi)}$$

Hence eqn (xv) becomes $\vec{H} = \frac{1}{\mu_0 c} (\hat{n} \times \vec{E})$

Taking modulus $|\vec{H}| = H = \frac{1}{\mu_0 c} |\hat{n} \times \vec{E}| = \frac{1}{\mu_0 c} E \sin 90^\circ = \frac{1}{\mu_0 c} E$

$$\text{Let } Z_0 = \left| \frac{\vec{E}}{\vec{H}} \right| = \frac{E}{H} = \mu_0 c = \sqrt{\frac{\mu_0}{\epsilon_0}} \quad \text{--- (xvii) } \left[\because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \right]$$

$$\frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{\frac{4\pi \times 10^{-7}}{8.85 \times 10^{-12}}} = 376.6 \text{ ohm}$$

$$\text{As } E \Rightarrow \frac{\text{V}}{\text{m}}, H \Rightarrow \frac{\text{Amp}}{\text{m}}, \therefore \frac{E}{H} \Rightarrow \frac{\text{V} \times \text{m}}{\text{Amp} \times \text{m}} = \frac{\text{Volt}}{\text{Amp}} = \text{ohm}$$

Thus unit of $\frac{E}{H}$ or Z_0 is same as that of resistance or impedance.

Hence Z_0 is called the wave impedance of free space.

As the ratio $\left| \frac{\vec{E}}{\vec{H}} \right| = \mu_0 c$ is real and +ve

or or $\left| \frac{\vec{E}_0}{\vec{H}_0} \right| = \mu_0 c$ " " " "

It implies that the field vectors $\vec{E}$ and $\vec{H}$ are in the same phase.

(28)

Plane e.m. waves in dielectric (or Non-conducting)Isotropic medium:

In an isotropic dielectric medium, the current density $\vec{J}$ and volume charge density ρ are zero. Further, the displacement vector $\vec{D}$ and magnetic field $\vec{B}$ are defined as $\vec{D} = \epsilon \vec{E}$ and $\vec{B} = \mu \vec{H}$. In fact $\vec{D} = \epsilon_0 \vec{E} + \vec{P} \equiv \epsilon \vec{E}$ and $\vec{B} = \mu_0 \vec{H} + \vec{M} \equiv \mu \vec{H}$.

For isotropic dielectric medium

$$\vec{D} = \epsilon \vec{E}, \quad \vec{B} = \mu \vec{H}, \quad \vec{J} = \sigma \vec{E} = 0 \text{ and volume charge density } \rho = 0$$

Hence, Maxwell's Equations becomes

$$\vec{\nabla} \cdot \vec{E} = 0 \quad \text{--- (i)} \quad \vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t} \quad \text{--- (ii)}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{--- (iii)} \quad \vec{\nabla} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t} \quad \text{--- (iv)}$$

Taking curl of Eq. (ii), we get

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left[\mu \frac{\partial \vec{H}}{\partial t} \right]$$

$$\text{or } \nabla (\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{H}) = -\mu \frac{\partial}{\partial t} \left(\epsilon \frac{\partial \vec{E}}{\partial t} \right)$$

$$0 - \nabla^2 \vec{E} = -\mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\text{or } \boxed{\nabla^2 \vec{E} = \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}} \quad \text{--- (v)}$$

Similarly, taking curl of equation (iv) and by using eqs (i) & (iii), then, we get

$$\boxed{\nabla^2 \vec{H} = \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2}} \quad \text{--- (vi)}$$

As discuss earlier $1/\sqrt{\mu\epsilon}$ gives the phase velocity v of the wave in the medium. Then

$$\nabla^2 \vec{E} - \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0 \quad \text{--- (vii)} \quad \text{where } v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$$

$$\text{and } \nabla^2 \vec{H} - \frac{1}{v^2} \frac{\partial^2 \vec{H}}{\partial t^2} = 0 \quad \text{--- (viii)}$$

$$v = \frac{1}{\sqrt{\mu_r \epsilon_r \mu_0 \epsilon_0}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

The plane wave solution of wave equations (vii) & (viii) are

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{--- (ix)}$$

$$\text{and } \vec{H}(\vec{r}, t) = \vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{--- (x)}$$

where $\vec{E}_0$ and $\vec{H}_0$ are complex amplitudes which are constants in space and time and $\vec{k}$ is the wave propagation vector;

$$\vec{k} = k \hat{n} = \frac{2\pi}{\lambda} \hat{n} = \frac{2\pi}{\lambda} \times \frac{v}{v} \hat{n} = \frac{2\pi v}{\lambda v} \hat{n} = \frac{\omega}{v} \hat{n} = \frac{\omega}{c} \hat{n}$$

$$\text{or } \vec{k} = \frac{\omega}{v} \hat{n} \quad \text{--- (xi)}$$

Here, $\hat{n}$ is a unit vector in the direction of wave propagation.

Relative directions of $\vec{E}$ and $\vec{H}$

By Maxwell's Ist Equation

$$\vec{\nabla} \cdot \vec{E} = 0 \Rightarrow i\vec{k} \cdot \vec{E} = 0 \quad \text{or } \vec{k} \cdot \vec{E} = 0$$

$$\text{and } \vec{\nabla} \cdot \vec{H} = 0 \Rightarrow i\vec{k} \cdot \vec{H} = 0 \quad \text{or } \vec{k} \cdot \vec{H} = 0$$

Hence both $\vec{E}$ and $\vec{H}$ vectors are perpendicular to $\vec{k}$.

Thus e.m. waves are transverse waves.

Also from the IIIrd Maxwell's equation

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\text{or } i\vec{k} \times \vec{E} = -\mu(-i\omega \vec{H})$$

$$\text{or } \vec{k} \times \vec{E} = \mu\omega \vec{H} \quad \text{--- (xii)}$$

Hence, from eqn (xii) it is clear $\vec{H}$ is $\perp$ to $\vec{k}$ and $\vec{H}$ is $\perp$ to $\vec{E}$

Also from IVth Maxwell's eqn.

$$\vec{\nabla} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$i\vec{k} \times \vec{H} = \epsilon(-i\omega \vec{E})$$

$$\text{or } \vec{k} \times \vec{H} = -\epsilon\omega \vec{E}$$

$$\text{or } \vec{H} \times \vec{k} = \epsilon\omega \vec{E} \quad \text{--- (xiii)}$$

Equations (xii) and (xiii) reveal that $\vec{E}$, $\vec{H}$ & $\vec{k}$ form a right-handed system of coordinate axis

Phase of $\vec{E}$ and $\vec{H}$ and wave Impedance of dielectric medium:

From eqn (xii)

$$\vec{k} \times \vec{E} = \mu\omega \vec{H} \quad \text{or } \vec{H} = \frac{1}{\mu\omega} \vec{k} \times \vec{E}$$

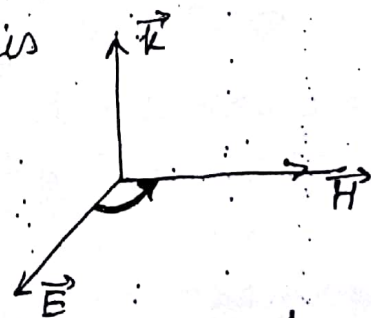
$$\text{or } \vec{H} = \frac{1}{\mu\omega} (k \hat{n} \times \vec{E}) = \frac{k}{\mu\omega} (\hat{n} \times \vec{E}) = \frac{\omega}{v} \cdot \frac{1}{\mu\omega} (\hat{n} \times \vec{E})$$

$$\vec{H} = \frac{1}{\mu v} (\hat{n} \times \vec{E}) = \sqrt{\frac{\epsilon}{\mu}} (\hat{n} \times \vec{E})$$

$$\text{Taking modulus of } |\vec{H}| = H = \sqrt{\frac{\epsilon}{\mu}} E \sin 90^\circ = \sqrt{\frac{\epsilon}{\mu}} E$$

$$\therefore \text{Impedance } Z = \frac{E}{H} = \frac{E}{\sqrt{\frac{\epsilon}{\mu}} E} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_r \mu_0}{\epsilon_r \epsilon_0}} = \sqrt{\frac{\mu_r}{\epsilon_r}} \cdot Z_0$$

$\therefore \vec{E}$ & $\vec{H}$ have same phase.



$$\begin{aligned} \therefore k &= \frac{\omega}{v} \\ v &= \frac{1}{\sqrt{\mu\epsilon}} \end{aligned}$$

E.M. Waves in conducting medium

We consider a conducting linear and isotropic medium whose permeability is μ , permittivity is ϵ and the conductivity is σ . Further, we assume that there is no free charge available inside the conductor, i.e. $\rho = 0$. Under this situation, we can write the Maxwell's equation as

$$\vec{\nabla} \cdot \vec{E} = 0 \quad \text{--- (1)}$$

$$\vec{\nabla} \cdot \vec{H} = 0 \quad \text{--- (2)}$$

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t} \quad \text{--- (3)}$$

$$\vec{\nabla} \times \vec{H} = \vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \quad \text{--- (4)}$$

$$\text{or } \vec{\nabla} \times \vec{H} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \quad \text{--- (4)} \quad \boxed{J = \sigma E}$$

Taking curl of eq (3), we have

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla} \times \left[-\mu \frac{\partial \vec{H}}{\partial t} \right] = -\mu \frac{\partial}{\partial t} [\vec{\nabla} \times \vec{H}]$$

$$= -\mu \frac{\partial}{\partial t} \left[\sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t} \right]$$

$$\vec{\nabla} (\vec{\nabla} \cdot \vec{E}) - (\vec{\nabla} \cdot \vec{\nabla}) \vec{E} = -\mu \sigma \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla} \cdot \vec{E} = 0, \text{ so } -\nabla^2 \vec{E} = -\mu \sigma \frac{\partial \vec{E}}{\partial t} - \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\text{or } \boxed{\nabla^2 \vec{E} = \mu \sigma \frac{\partial \vec{E}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{E}}{\partial t^2}} \quad \text{--- (5)}$$

Eq (5) is the electromagnetic wave equation for the electric field in the conducting medium.

Similarly we can obtain electromagnetic wave eqn for magnetic field vector in conducting medium, i.e.

$$\nabla^2 \vec{H} = \mu \sigma \frac{\partial \vec{H}}{\partial t} + \mu \epsilon \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{--- (6)}$$

Plane wave solutions of eqs (5) and (6)

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{--- (7)}$$

$$\vec{H}(\vec{r}, t) = \vec{H}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \quad \text{--- (8)}$$

Differentiating eqn (7) w.r. to t twice, then

$$\frac{\partial \vec{E}}{\partial t} = (-i\omega) \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\frac{\partial^2 \vec{E}}{\partial t^2} = (-i\omega) \times (-i\omega) \vec{E}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} = -\omega^2 \vec{E}$$

$$\boxed{\frac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2 \vec{E}}$$

$$\text{Also } \nabla^2 \vec{E} = (\vec{\nabla} \cdot \vec{\nabla}) \vec{E} = (i\vec{k} \cdot i\vec{k}) \vec{E} = -k^2 \vec{E}$$

$$\boxed{\nabla^2 \vec{E} = -k^2 \vec{E}}$$

(31) Putting these values in eqn (5); then

$$-k^2 \vec{E} - \sigma \mu (-i\omega \vec{E}) - \epsilon \mu (-\omega^2 \vec{E}) = 0$$

$$-(k^2 - \sigma \mu \omega i - \epsilon \mu \omega^2) \vec{E} = 0$$

Since $\vec{E}$ is arbitrary

$$k^2 - \epsilon \mu \omega^2 - i \sigma \mu \omega = 0$$

$$\text{or } k^2 = \epsilon \mu \omega^2 + i \sigma \mu \omega \quad \text{--- (9)}$$

$$\text{or } \boxed{k^2 = \epsilon \mu \omega^2 \left(1 + \frac{i \sigma}{\epsilon \omega}\right)} \quad \text{--- (9)}$$

Here k is a complex and it is the relation b/w k and ω . Hence it is a dispersion relation. $k = \alpha + \beta i$ with this

Let $\vec{k} = k \hat{n}$, $\hat{n}$ = a unit vector in the direction of wave propagation

$$k = \text{wave number} = \frac{2\pi}{\lambda}$$

k is a complex, let

$$k = \alpha + i\beta$$

on squaring

$$k^2 = \alpha^2 - \beta^2 + 2i\alpha\beta \quad \text{--- (10)}$$

Comparing eqn (10) and eqn (9), we get

$$\alpha^2 - \beta^2 = \epsilon \mu \omega^2 \quad \text{and} \quad 2\alpha\beta = \sigma \mu \omega$$

$$\text{or } \alpha = \frac{\sigma \mu \omega}{2\alpha} \quad \beta = \frac{\sigma \mu \omega}{2\alpha}$$

$$\alpha^2 - \left(\frac{\sigma \mu \omega}{2\alpha}\right)^2 - \epsilon \mu \omega^2 = 0$$

$$4\alpha^4 - 4\epsilon \mu \omega^2 \alpha^2 - (\sigma \mu \omega)^2 = 0$$

by using relation $ax^2 + bx + c = 0$

on solving, we get α and β

$$\alpha = \omega \sqrt{\frac{\epsilon \mu}{2}} \left[1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon \omega}\right)^2}\right]^{1/2} \quad \text{--- (11)}$$

$$\beta = \omega \sqrt{\frac{\epsilon \mu}{2}} \left[-1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon \omega}\right)^2}\right]^{1/2} \quad \text{--- (12)}$$

Now putting $k = (\alpha + i\beta)$ and $\vec{k} = k \hat{n}$ in eqs (7) and (8), we get

$$\text{or } \vec{E}(\vec{r}, t) = \vec{E}_0 e^{i(\alpha \hat{n} \cdot \vec{r} - \omega t)} e^{-\beta \hat{n} \cdot \vec{r}} \quad \text{--- (13)}$$

and $\vec{H}(\vec{r}, t) = \vec{H}_0 e^{i\{\alpha + i\beta\} \hat{n} \cdot \vec{r} - \omega t}$
 $\vec{H}(\vec{r}, t) = \vec{H}_0 e^{-\beta \hat{n} \cdot \vec{r}} e^{i(\alpha \hat{n} \cdot \vec{r} - \omega t)} \quad (14)$

The eqs (13) and (14) show that the field amplitudes are attenuated spatially due to the factor $e^{-\beta \hat{n} \cdot \vec{r}}$. Thus, the amplitude decreases with increase of β and $\vec{r}$.

Why is attenuation? : As the oscillatory fields $\vec{E}$ and $\vec{H}$ vary in magnitudes, they establish currents in the medium, the current generates heat and thus energy is dissipated. It causes slow dissipation and attenuation. Here β is called absorption coefficient.

In the field vector equations (13) and (14), the k notation has been replaced by α . The phase velocity of the wave

$$v = \frac{\omega}{k} = \frac{\omega}{\alpha} = \frac{1}{\sqrt{\mu\epsilon} \left[1 + \sqrt{1 + \left(\frac{\sigma}{\epsilon\omega} \right)^2} \right]^{1/2}}$$

wavelength $\lambda = \frac{v}{\nu} = \frac{v \cdot 2\pi}{\nu \cdot 2\pi} = \frac{2\pi v}{2\pi \nu} = \frac{2\pi v}{\omega} = \frac{2\pi}{\omega/v}$

$$\lambda = \frac{2\pi}{\alpha}$$

Phases of $\vec{E}$ & $\vec{H}$: Since $\vec{k} \times \vec{E} = \mu\omega \vec{H}$
 $\therefore \vec{H} = \frac{1}{\mu\omega} (\vec{k} \times \vec{E}) = \frac{1}{\mu\omega} (\hat{n} k \times \vec{E})$
 $\vec{H} = \frac{k}{\mu\omega} (\hat{n} \times \vec{E}) = \frac{1}{\mu\omega} (\alpha + i\beta) (\hat{n} \times \vec{E})$

Modulus of $\vec{H} = |\vec{H}| = H = \frac{1}{\mu\omega} (\alpha + i\beta) E$ Since $\hat{n} \times \vec{E} = E \sin 90^\circ = E$

$\therefore \frac{H}{E} = \left| \frac{\vec{H}}{\vec{E}} \right| = \frac{H_0}{E_0} = \frac{\alpha + i\beta}{\mu\omega} = \text{A complex number}$

Thus $\vec{E}$ and $\vec{H}$ vectors are not in phase in a conducting medium.

Skin Depth or Penetration Depth : From the eqs (13) and (14) the factor $e^{-\beta \hat{n} \cdot \vec{r}}$ shows attenuation of wave function with depth. If $z = 1/\beta$, then $e^{-1} = \frac{1}{e} = 0.369$ is called the skin depth say δ . The skin depth is decided by the imaginary part of the wave vector i.e. β

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Skin Depth: The wave given by Eq. (14) shows an exponential damping and attenuation with distance. Greater is the value of β , greater is the attenuation. The term $(1/\beta)$ measures the depth at which electromagnetic wave entering a conductor is damped to $1/e = 0.369$ of its amplitudes at the surface. The depth is known as skin depth or the penetration depth and is usually represented by δ .

$$\delta = \frac{1}{\beta} = \frac{1}{\omega \sqrt{\mu \epsilon} \left[\frac{-1 + \sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2}}{2} \right]^{-1/2}}$$

$$= \sqrt{\left(\frac{2}{\mu \sigma \omega} \right)}$$

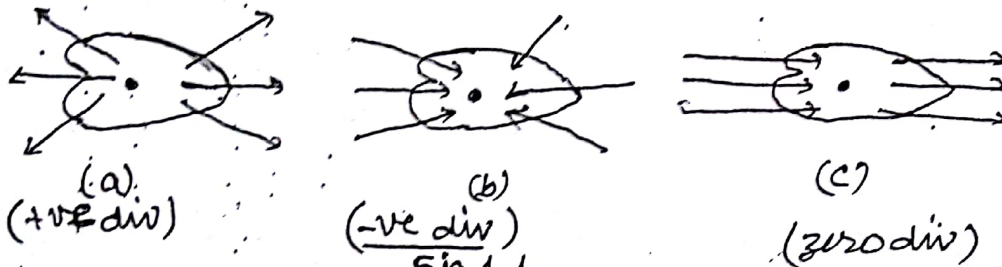
For good conductor it is characterised by $\frac{\sigma}{\omega \epsilon} \gg 1$

Q1. Define the divergence of a vector and give its significance.

Ans: The divergence of a vector field at any point is defined as the amount of flux per unit volume diverging from that point.

The physical significance of divergence of a vector field is that at a point it gives the amount of flux per unit volume diverging from that point.

• If the divergence of a vector field at a point is



+ve, it means that the flux is coming out the surface around that point (Fig. 1.1(a)).

• If the divergence of a vector field is -ve, it means that the flux is entering through the surface (Fig. 1.1(b)).

• If the divergence of a vector field is zero means that the flux entering through the surface is equal to the flux leaving through that surface as shown in Fig. 1.1(c).

Q2. Define curl of a vector field.

Ans: (i) The rate of change of a vector field is also called curl, which means circular rotation.

(ii) The curl of a vector field is defined as the maximum line integral of vector per unit area.

Q3. What do you mean by (i) Solenoidal (ii) Irrotational?

Ans: (i) Solenoidal: A vector whose divergence is zero is called solenoidal.

(ii) Irrotational: A vector whose curl is zero is called irrotational.

Q4. What is lamellar field?

Ans: A vector field which can be expressed as the gradient of a scalar field is known as a lamellar (or laminae) vector field.

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Electric field is an example of a lamellar field, since

$$\mathbf{E} = -\nabla V$$

where V is the electrical potential, which is a scalar function.

The word lamellar means that the field can be divided into layers over which the value of scalar function whose gradient gives the vector field $\mathbf{E}$, remains constant.